INCH-POUND
MIL-PRF-94/5H
16 February 2001
SUPERSEDING
MIL-PRF-94/5G
26 October 1981

PERFORMANCE SPECIFICATION

RESISTORS, VARIABLE, COMPOSITION, STYLE RV4

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for style RV4 composition, variable resistors.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-94 - Resistor, Variable, Composition, General Specification for.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Service, Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.3 Order of precedence. In event of a conflict between the text of this document and the references cited herein (except for related associated specifications, specification sheets, or MS sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC-VAT, Post Office Box 3990, Columbus, OH 43216-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

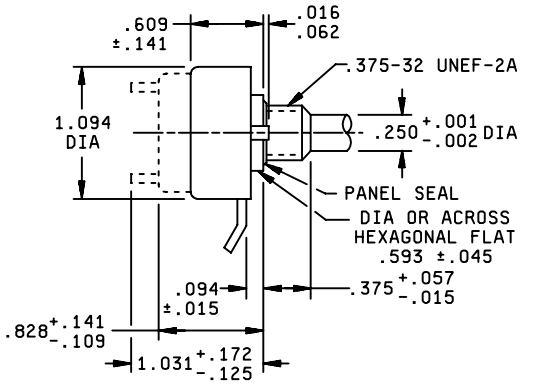
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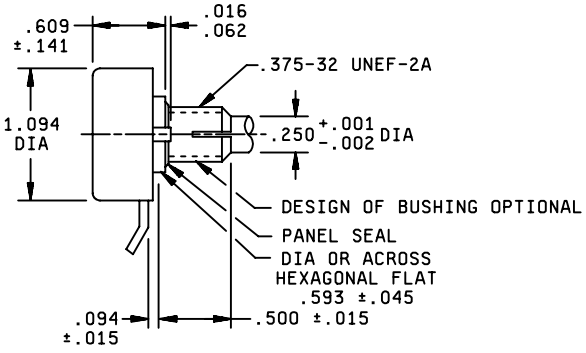
3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-94.

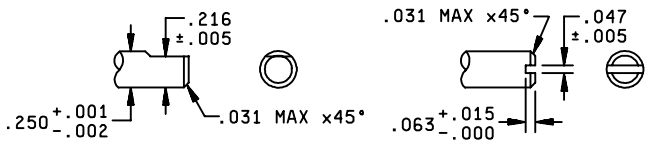
3.2 Interface and physical dimensions. The resistors shall meet the interface and physical dimensions specified in figure 1.



SHAFT-AND PANEL-SEALED-STANDARD BUSHING TYPE



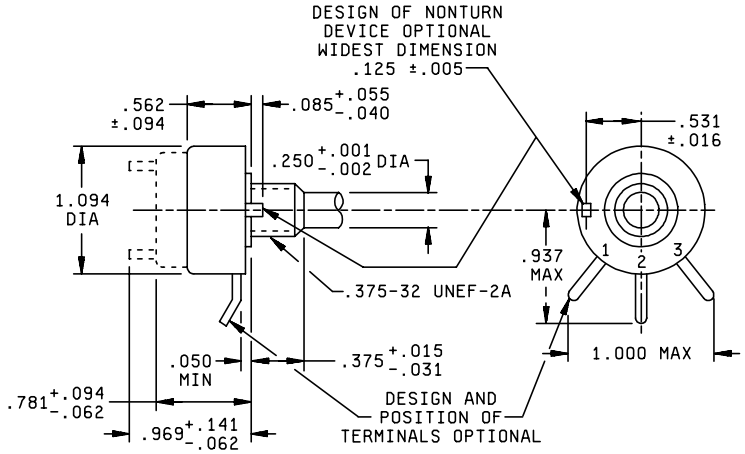
SHAFT-AND PANEL-SEALED-LOCKING BUSHING TYPE



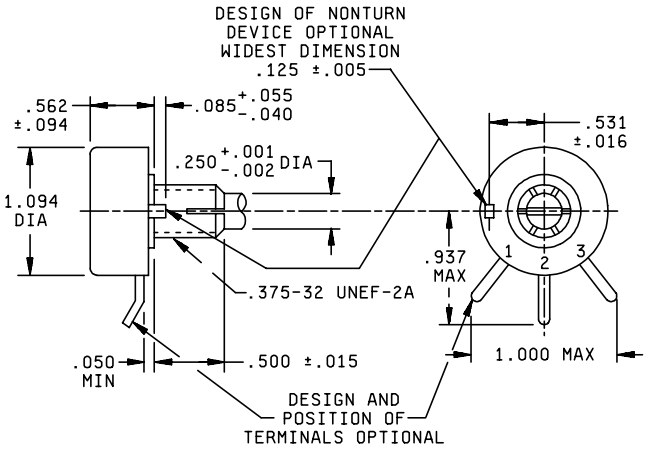
FLATTED SHAFT SLOTTED SHAFT

FIGURE 1. Style RV4.

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STANDARD-BUSHING TYPE



LOCKING-BUSHING TYPE

Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
0.001	0.03	0.031	0.79	0.063	1.60	0.172	4.37	0.593	15.06
0.002	0.05	0.045	1.14	0.094	2.39	0.216	5.49	0.609	15.47
0.005	0.13	0.047	1.19	0.109	2.77	0.250	6.35	0.828	21.03
0.015	0.38	0.057	1.45	0.125	3.18	0.375	9.53	1.031	26.19
0.016	0.41	0.062	1.57	0.141	3.58	0.500	12.70	1.094	27.79

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information.
3. Unless otherwise specified, tolerance is ±0.062 (1.57mm).

FIGURE 1. Style RV4 Continued.

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3.3 Nominal total resistance and rated continuous working voltages. The nominal total resistance and rated continuous working voltage (RCWV) shall be as specified in table I.

TABLE I. Nominal total resistance and rated continuous working voltage.

Nominal total resistance (in ohms)	RCWV 1/ (volts)		Nominal total resistance (in megohms)	RCWV 1/ (volts)	
	Taper A	Tapers C and F		Taper A	Tapers C and F
50	10		0.10	445	316
100	14	10	0.15	500	350
150	17	12	0.20 2/	500	350
200 2/	20		0.25	500	350
250	22	16	0.35	500	350
350	26	18	0.50	500	350
500	32	23	0.75	500	350
750	39	28	1.00	500	350
1,000	45	32	1.50	500	350
1,500 2/	55	39	2.00	500	350
2,000	63.3	45	2.50	500	350
2,500	71	50	3.50	500	350
3,500	84	60	5.00	500	350
5,000	100	71			
7,500	122	87			
10,000	141	100			
15,000	173	123			
20,000 2/	200	141			
25,000	224	160			
35,000	264	187			
50,000	316	224			
75,000	387	249			

1/ Rated for continuous working voltage at 70°C.
2/ For replacement purposes only. Not for new design.

3.4 Shaft length. The length of the operating shaft shall be in accordance with table II.

TABLE II Shaft length. 1/

Symbol	Length of shafts from mounting surface of resistors, (nominal)		
	Flatted	Slotted	
	Bushings N and S (inches ±0.313)	Bushings N and S (inches ±0.313)	Bushings L and T (inches ±0.313)
A			0.625
B		0.500	
D	0.875	0.875	0.875
G		1.250	
N		1.500	
J		2.000	
K		2.500	

1/ For resistors with panel seals, the mounting surface shall be considered the metal face of the panel seal when the seal is firmly seated against the resistor body.

3.5 Mounting and locking nuts. The mounting nut shall be 0.09375 inches thick and measure 0.500 inches across the hexagonal flats. The locking nut shall be within the limits 0.151 inches and 0.234 inches thick and shall measure 0.500 inches across the hexagonal flats. The thread size shall be 0.375-32 NEF-2B.

3.5.1 Internal tooth lockwasher. Internal tooth lockwasher of suitable size shall be applied.

3.5.2 Retainer rings. If retainer rings are used, they shall have a maximum thickness of 0.047 inches.

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3.6 Power rating. The power rating shall be 2.0 watts for taper A resistors and 1.0 watts for taper C and F.

3.7 Torque.

3.7.1 Operating torque. The torque required to effect rotation shall be 1 ounce-inch minimum and 6 ounce-inch maximum.

3.7.2 Stop torque. The torque applied to the operating shaft to the stops shall be 8 pound-inch.

3.7.3 Locking torque (as applicable). For the locking bushing type resistors, the locking nut shall be tightened with a torque of 10 pound-inches. After the locking nut is tightened, the contact arm shall not move when a torque of 30 ounce-inches is applied to the shaft.

3.8 Total mechanical rotation. The total mechanical rotation with a switch shall be within the limits of 330 degrees and 340 degrees; without a switch, within the limits of 309 degrees and 320 degrees.

3.9 Thermal cycling. The resistance shall not change in excess of 4 percent.

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-PRF-94.

4.2 Dielectric withstanding voltage. In the dielectric withstanding voltage test, the applied potential shall be 450 volts rms at reduced barometric pressure.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The intended use specified in MIL-PRF-94 is applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification, and the complete PIN (see 1.2).

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of the individual documents referenced (see 2.1).

c. Packaging instructions (see 5.1).

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Review activities
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Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-94/5H	**2. DOCUMENT DATE (YYMMDD)** **01/02/16**

3. DOCUMENT TITLE RESISTORS, VARIABLE, COMPOSITION, STYLE RV4

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Incl Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Supply Center, Columbus ATTN: DSCC/VAM	b. TELEPHONE (Include Area Code) (1) Commercial (614) 692-8754 (2) AUTOVON 850-8754
c. ADDRESS (Include Zip Code) 3990 East Broad Street Columbus, OH 43213-1199	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5803 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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